



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Instrumentation Engineering

In-Semester Examination I

Academic Year: 2025-26 (semester I)

Class: Third Year

Course: Automatic Control System

Course Code: PCC-IN303

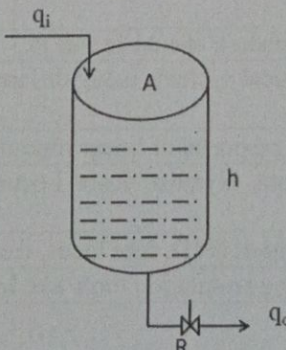
Date: 22.08.2025

Time: 03:00 to 03:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Question 1 compulsory and solve any three questions from Q.2, 3, 4 and 5.**

Q.N.		Marks	CO	BT
Q.1	Define the following terms i. Plant ii. Process iii. System iv. Manipulated variable v. Control variables	5	CO1	BT2
Q.2	Explain the open loop and closed loop systems with block diagram and examples.	5	CO1	BT2
Q.3	Derive a TF model using mass balance for following single cylindrical liquid level process.  Where, q_i = input flow or input variable, h = liquid level or output variable, A = cross section of tank, R = valve resistance, q_o = outflow.	5	CO1	BT3
Q.4	Simplify the following block diagram using block reduction technique	5	CO1	BT3

Q.5	Develop a TF model for following signal flow graph using Mason's Gain Formula	5	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Develop mathematical models of physical systems using differential equations and Laplace transforms.
CO 2	Analyze the transient and steady-state response of linear time-invariant systems.
CO 3	Apply stability criteria (Routh-Hurwitz, Nyquist, Root Locus, and Bode) to evaluate system stability.
CO 4	Design controllers using classical methods (PID, root locus, frequency response).
CO 5	Implement control system designs using simulation tools like MATLAB and Simulink.



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Department of Instrumentation Engineering

In-Semester Examination I

Academic Year: 2025-26 (semester I)

Class: Third Year

Course: Feedback Control System

Course Code: PCC-IN301

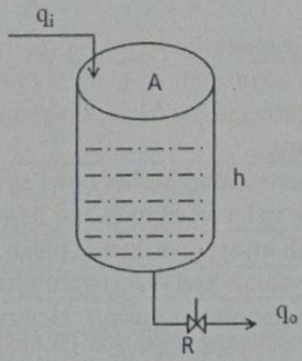
Date: 22.08.2025

Time: 03:00 to 03:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. **Question 1 compulsory and solve any three questions from Q.2, 3, 4 and 5.**

Q.N.		Marks	CO	BT
Q.1	Define the following terms i. Plant ii. Process iii. System iv. Manipulated variable v. Control variables	5	CO1	BT2
Q.2	Explain the open loop and closed loop systems with block diagram and examples.	5	CO1	BT2
Q.3	Derive a TF model using mass balance for following single cylindrical liquid level process.  Where, q_i =input flow or input variable, h = liquid level or output variable, A = cross section of tank, R = valve resistance, q_o = outflow.	5	CO1	BT3
Q.4	Simplify the following block diagram using block reduction technique	5	CO1	BT3

Q.5	Develop a TF model for following signal flow graph using Mason's Gain Formula	5	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Exhibit the capability to represent the mathematical model of physical systems using linear differential equations; Laplace transform and use block diagram algebra, Mason's gain formula to simplify complicated control systems.
CO 2	To determine time response of first, second and higher order systems to standard test signals and to specify control system performance in terms of time and frequency domain specifications.
CO 3	Construct Hurwitz determinant, Routh array, root-locus, polar plot, Bode plot and Nyquist plot for single variable feedback control systems and investigate system stability.
CO 4	Realize lead, lag and lag lead compensators using electrical, electronic and mechanical components.
CO 5	Validate the concepts of time domain, frequency domain and stability analysis using MATLAB.
CO 6	Gain some practical experience in control engineering which might become a future research point for them. (Some is not specific and measurable)



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
In-Semester Examination - I
Academic Year: 2025-26 (Semester – I)

Class: TY B.Tech.	Subject: Unit Operation & Instrumentation	Subject Code: PEC-IN307
Date: 18.08.2025	Time: 3:00 to 3:45	Max. Marks: 20

Note:

- i. Q.1 is compulsory. Solve any Two questions from Q.2, Q.3, and Q.4.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/Sharing of stationery, calculator, etc., not allowed.
- v. Figures to the right indicate full marks.
- vi. The symbols carry their usual meaning.

Q.N.			Marks	CO	BT
Q.1		State whether the following statements are True or False. Justify your answer. (Any 4) 1. A digital twin is a virtual replica of a physical system. 2. Predictive maintenance is not possible in Automation 4.0. 3. All equipments with their tags must be shown in PFD (Process Flow Diagram). 4. Batch distillation is characterised by its steady state operation. 5. Edge computer is a remote server.	6	CO1 CO2 & CO3	BT1, BT2 & BT4
Q.2	a	Explain 'Reversible Processes' and 'Irreversible Processes' with examples.	3	CO1	BT1 BT2
	b	Differentiate between PFD (Process Flow Diagram) and P&ID (Piping and Instrumentation Diagram).	4	CO2	BT1 BT2
Q.3	a	What are different types of Evaporators? Explain Force circulation evaporator with diagram.	3	CO4	BT1 BT2
	b	What is distillation process? Explain flash distillation with diagram.	4	CO4	BT3
Q.4	a	Enlist key liquid characteristics affecting evaporation and distillation with their impacts.	4	CO2	BT1 BT2
	b	Write a note on 'Sustainability in Chemical Processes'.	3	CO1	BT2 BT4

Bloom's Taxonomy (BT): BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the concept of unit and unit operation.
CO 2	Recognise the basic operations of different chemical processes.
CO 3	Know the constructions / mechanisms of different equipments used for unit operations.
CO 4	Use the knowledge of constructions and working of different equipments in studying the operations of different mass transfer and heat transfer operations.
CO 5	Select the process equipment for particular unit operation.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Instrumentation Engineering In-Semester Examination I Academic Year: 2025-26 (Semester – I)	
Class: TY B.Tech	Subject: Microprocessor & Microcontroller	Subject Code: PCC-IN 305
Date: 19.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Non Programmable calculator is allowed.
- v. Figures to the right indicate full marks.
- vi. The symbols carry the usual meaning.

Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	Differentiate between Microprocessor and Microcontroller.	4	CO1	BT1
Q.2	Explain in detail internal RAM of 8051 microcontroller.	4	CO1	BT2
Q.3	Describe following addressing modes(8051) with example a) Implied Addressing Mode b) Indexed Addressing Mode	4	CO1	BT2
Q.4	Explain TCON and TMOD registers in 8051 microcontroller.	4	CO1	BT1
Q.5	Describe functions of following pins in 8051 microcontroller. a) EA b) INT0 3) PSEN 4) ALE	4	CO1	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To describe basics of 8085, 8051 and its instruction set.
CO2	To understand historical development of microcontrollers and to know different 8, 16, 32 bit microcontrollers.
CO3	To solve assembly language programs based on the instruction set of 8085 and 8051.
CO4	To get insight of 8051 based hardware system and so to study ADC, keyboard etc.
CO5	To execute assembly language programs based on the instruction set of 8051

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Instrumentation Engineering In-Semester Examination I Academic Year : 2025 - 26 (Semester – I)	
Class: TY B.Tech	Subject: Industrial Instrumentation	Subject Code: PCC - IN301 / IN302
Date: 20.08.2025	Time: 15:00 to 15:45	Max. Marks : 20

Note:

- Assume suitable data whenever necessary and state it clearly.*
- Mobile Phones and Programmable Calculators are strictly prohibited.*
- Exchange / Sharing of Stationary, Calculator, etc. is not allowed.*
- Figures to the right indicate Full Marks.*
- The symbols carry the usual meaning.*
- Solve **ANY FOUR** Questions.*

Q. N.		Marks	CO	BT
Q. 1	Explain the sensors used for Temperature measurement.	05	CO1	BT2
Q. 2	Elaborate in detail, RTDs & Thermocouples.	05	CO1	BT2
Q. 3	Describe in detail, how Pressure is being measured in Industries.	05	CO2	BT2
Q. 4	Describe Flapper - Nozzle and I to P Converter.	05	CO3	BT2
Q. 5 a	Explain Air Filter Regulators.	03	CO3	BT2
b	What do you know about SMART Temperature Transducers.	02	CO1	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT):

BT1 – Remember, BT2 – Understand, BT3 – Apply, BT4 – Analyze, BT5 – Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall knowledge of Temperature Transducers like Thermocouples, Thermistors, RTDs, Pyrometers and other Temperature measuring techniques.
CO 2	Understand the construction and working principle of various transducers / sensors to measure Physical Quantities.
CO 3	Apply adequate knowledge of Pressure and Strain Transducers.
CO 4	Analyse, formulate and select suitable Sensors / Transducers for given Industrial Applications.
CO 5	Demonstrate working knowledge of Safety Practices used in Measurement and Control of Industrial Processes.